

Diversified but Crowded: Evidence from 13F Holdings

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Abstract

This paper examines whether diversification within institutional portfolios is associated with greater portfolio distinctiveness or greater overlap across managers. Using quarterly SEC 13F holdings for the 500 largest institutional managers from 2018Q1 to 2025Q4, the analysis constructs pairwise portfolio overlap and a manager-level measure of crowding. Institutional crowding rises substantially over the sample period, from 0.140 to 0.231. Within-quarter sorts show that average crowding increases from 0.051 in the lowest entropy quintile to 0.248 in the highest. Regression estimates show that more diversified portfolios are more crowded after controlling for portfolio characteristics and quarter fixed effects. Holdings-based crowding is also associated with stronger next-quarter common movement between manager portfolios and leave-one-out peer portfolios, including after adjusting for broad market and Fama–French factor exposure. This association remains stable after excluding the Big Three and other large managers. Holdings-based crowding also predicts stronger next-quarter common movement between manager portfolios and leave-one-out peer portfolios, including after adjusting for broad market and Fama–French factor exposure. Overall, the results show that diversification within individual portfolios does not necessarily imply distinctiveness across institutional investors.

Keywords: Institutional investors, SEC 13F filings, Portfolio crowding, Portfolio diversification, Portfolio overlap, Common ownership

1. Introduction

Diversification is a central principle of portfolio choice because spreading wealth across securities can reduce exposure to firm-specific risk (Markowitz, 1952). This conventional argument, however, concerns diversification within an individual portfolio. It does not imply that the portfolios of different investors are distinct from one another. Institutional managers may each hold a broad and individually diversified portfolio while concentrating collectively in the same liquid, index-relevant, or benchmark-compatible securities. This distinction has become increasingly important as index investing, benchmark-sensitive asset management, and the concentration of institutional assets have reshaped portfolio demand (Coles et al., 2022; Chinco and Sammon, 2024; Haddad et al., 2021; Jiang et al., 2025). Diversification at the manager level may therefore coexist with, rather than prevent, similarity across managers.

A large literature shows why such similarity may have economic consequences. Institutional investors frequently trade in the same direction (Sias, 2004), while flow-induced purchases and sales can generate price pressure in securities held by multiple funds (Coval and Stafford, 2007). Common ownership can expose securities to correlated non-fundamental demand shocks and greater price fragility (Greenwood and Thesmar, 2011). Stocks connected through shared institutional owners also exhibit stronger return comovement (Anton and Polk, 2014). More recent evidence shows that large institutional investors can exert substantial price effects when investment decisions are correlated across organizational units and benchmarks (Ben-David et al., 2021; Buffa and Hodor, 2023). Together, these studies show that common institutional positions can matter for prices, comovement, and financial fragility. They do not, however, directly answer whether managers that are more diversified within their own portfolios are also more similar to other institutional managers.

Several recent studies are more directly related to this question. Diversification requirements in the leveraged-loan market can induce institutional portfolios to become more similar (Elkamhi and Nozawa, 2022). Benchmarking can likewise affect institutional demand and asset prices (Pavlova and Sikorskaya, 2023). At the fund level, holdings overlap has been used to measure international mutual-fund crowding (Gonzalez et al., 2024), while entropy-based measures have recently been proposed to identify investor crowding (Perras and Wagner, 2026).

This paper is related to these studies but focuses on a narrower cross-manager question: whether broader diversification among U.S. institutional equity managers is systematically associated with greater portfolio overlap. The contribution is therefore not a new model of crowding or a causal estimate of shock transmission. It is a short empirical regularity: portfolios that are more diversified within managers can also be less distinct across managers.

2. Data and methodology

The sample comes from quarterly SEC 13F holdings from 2018Q1 to 2025Q4. In each quarter, the analysis retains the top 500 managers by reported portfolio value and recomputes portfolio weights within that manager universe. After cleaning, duplicate removal, and quarter-level consistency checks, the manager-level panel contains 16,000 manager-quarter observations. The holdings panel contains 17,147,528 manager-stock-quarter observations, the pairwise overlap panel contains 3,787,710 manager-pair-quarter observations, and the stock-level panel contains 400,564 stock-quarter observations.

The measurement pipeline is: 13F holdings $\rightarrow$ manager-stock weights $\rightarrow$ pairwise overlap $\rightarrow$ manager-level crowding $\rightarrow$ diversification measures $\rightarrow$ regressions and common portfolio movement validation. For manager m , stock s , and quarter t , let $w_{m,s,t}$ denote the portfolio weight. Pairwise overlap is

$$Overlap_{mn,t} = \sum_{s \in S_t} \min(w_{m,s,t}, w_{n,s,t}). \quad (1)$$

This is a holdings-based measure of cross-manager portfolio similarity. Manager-level crowding is the average leave-one-out overlap with peer managers:

$$Crowding_{m,t} = \frac{1}{N_t - 1} \sum_{n=1}^{N_t} \mathbf{1}_{\{n \neq m\}} Overlap_{mn,t}. \quad (2)$$

The indicator excludes manager m from its own peer average.

The baseline diversification measure is entropy,

$$Entropy_{m,t} = - \sum_s w_{m,s,t} \ln(w_{m,s,t}), \quad (3)$$

where higher values indicate broader diversification. Alternative measures are the Herfindahl–Hirschman index, $HHI_{m,t} = \sum_s w_{m,s,t}^2$, and effective holdings, $EffectiveN_{m,t} = 1/HHI_{m,t}$.

The analysis also uses abnormal crowding as a mechanical-breadth diagnostic: observed crowding minus expected random overlap for portfolios with the same number of holdings in the same quarter.

The baseline regression estimates the association between manager-level crowding and portfolio diversification:

$$Crowding_{m,t} = \alpha + \beta Diversification_{m,t} + X'_{m,t} \gamma + \delta_t + \varepsilon_{m,t}. \quad (4)$$

The control vector $X_{m,t}$ includes number of holdings, log portfolio value, and maximum position. All baseline specifications include quarter fixed effects, and standard errors are clustered by manager. The regression sample excludes manager-quarters with five or fewer holdings, effective holdings above 400, or a maximum portfolio position of at least 95%. These screens remove small or mechanically unusual portfolios. The resulting baseline regression sample contains 15,576 manager-quarter observations.

To assess economic relevance, approximate next-quarter daily manager returns are constructed from the top 300 value-ranked securities with available daily prices. The peer portfolio return is leave-one-out: for each manager, it is the equal-weighted average return of other managers in the same quarter, so the manager's own return cannot mechanically enter its peer benchmark.



Figure 1: Institutional portfolio crowding over time

Table 1: Within-quarter quintile sorts by portfolio entropy

Entropy quintile	Crowding mean	Holdings	Entropy	Effective N	HHI
1	0.0515	95.1	2.336	10.58	0.2306
2	0.1136	325.8	3.867	32.02	0.0388
3	0.1782	667.0	4.732	61.65	0.0191
4	0.2353	1344.4	5.445	103.52	0.0110
5	0.2476	2926.2	6.116	181.49	0.0066

3. Empirical results

3.1. Diversification and crowding

The average manager holds 1,072 securities, has entropy of 4.499, and has portfolio overlap of 0.165 with peer managers. Manager-level crowding rises from 0.140 in 2018Q1 to 0.231 in 2025Q4. Figure 1 shows this upward pattern.

Managers are sorted into entropy quintiles separately within each quarter. Table 1 reports the corresponding averages. Entropy measures portfolio breadth, HHI measures concentration, effective $N = 1/HHI$, and crowding is average leave-one-out pairwise overlap with peer managers. Average crowding rises from 0.051 in the lowest entropy quintile to 0.248 in the highest entropy quintile. Thus, managers with broader and less concentrated portfolios are also more similar to other institutional managers.

Table 2, Panel A shows that the diversification–crowding association remains after adding portfolio controls and quarter fixed effects. Entropy and effective holdings enter positively, while HHI enters negatively, with all three measures pointing in the same direction. Panel B is discussed in Section 3.3.

Table 2: Diversification regressions and mega-manager robustness

Panel A: Diversification regressions				
Dependent variable	(1)	(2)	(3)	
Focal variable	Crowding mean Entropy	Crowding mean Effective N	Crowding mean HHI	
Entropy	0.0657*** (0.0031)			
Effective N		0.0001*** (0.0001)		
HHI			-0.2002** (0.0833)	
Portfolio controls	Yes	Yes	Yes	
Quarter fixed effects	Yes	Yes	Yes	
Manager-clustered SE	Yes	Yes	Yes	
Observations	15,576	15,576	15,576	
R^2	0.645	0.489	0.486	
Panel B: Big Three and mega-manager robustness				
Sample	Entropy coef.	SE	Observations	R^2
Baseline	0.0657***	(0.0031)	15,576	0.645
Exclude Big Three	0.0652***	(0.0031)	15,476	0.641
Exclude top 5 managers	0.0629***	(0.0031)	15,067	0.635
Exclude top 10 managers	0.0614***	(0.0032)	14,609	0.629

Panel A reports OLS estimates with portfolio controls, quarter fixed effects, and manager-clustered standard errors. Effective $N = 1/HHI$.

Panel B reconstructs the portfolio-overlap network after each manager exclusion before re-estimating the baseline entropy specification.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

3.2. Common portfolio movement

Manager-level crowding has economic relevance if overlapping institutional portfolios subsequently move together. Figure 2 shows a monotonic pattern: future common return correlation rises from 0.737 in the lowest crowding quintile to 0.991 in the highest crowding quintile.

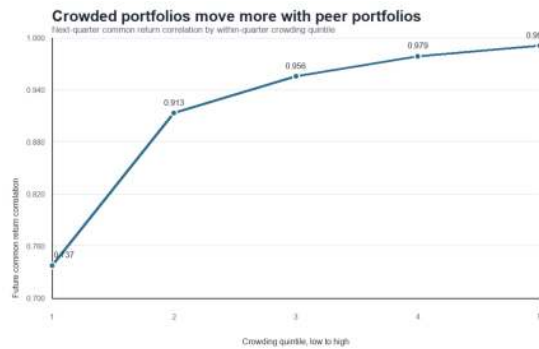


Figure 2: Crowding and subsequent common portfolio movement

Table 3 links observed and abnormal crowding to next-quarter common portfolio movement. Panel A shows positive associations with raw common return correlation. Panels B and C repeat the test after residualizing manager

and peer returns with respect to SPY and daily Fama–French factors. The positive residual-correlation estimates indicate that the crowding measure is related to economically relevant common movement, but the estimates should not be interpreted as causal effects.

Table 3: Crowding and subsequent common portfolio movement

Outcome	Focal variable	Coef.	SE	Observations
Panel A: Raw common portfolio movement				
Common corr.	Crowding mean	0.682***	(0.049)	6,395
Common corr.	Abnormal crowding	0.672***	(0.052)	6,395
Panel B: SPY-residual common portfolio movement				
SPY residual corr.	Crowding mean	0.960***	(0.067)	6,395
SPY residual corr.	Abnormal crowding	0.983***	(0.069)	6,395
Panel C: Fama-French residual common portfolio movement				
FF3 residual corr.	Crowding mean	0.822***	(0.074)	6,407
FF3 residual corr.	Abnormal crowding	0.835***	(0.076)	6,407

Outcomes are next-quarter correlations between manager returns and leave-one-out peer returns. All models include entropy, log portfolio value, max position, number of holdings, and quarter fixed effects; standard errors are clustered by manager.

Abnormal crowding subtracts a same-quarter random holding-count benchmark from observed crowding. Panels B and C use SPY- and Fama–French three-factor residualized returns. These specifications are diagnostic, not causal.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

3.3. Robustness

A central concern is that the diversification–crowding relation may reflect index giants or other mega managers. Table 2, Panel B addresses this concern by recomputing the portfolio-overlap network after excluding the Big Three and the largest managers, then re-estimating the baseline entropy specification. The entropy coefficient remains positive and close to the baseline estimate, including after excluding the top 10 managers. This stability indicates that the main association is not solely driven by the largest institutions in the 13F universe.

A second concern is mechanical overlap from broader portfolios. The main regressions control for number of holdings, while Table 3 also uses abnormal crowding, which subtracts a same-quarter random holding-count benchmark from observed crowding. Appendix Table A.2 shows that the raw common portfolio movement estimates are stable across priced-security universes. These tests support the descriptive interpretation: diversification is systematically related to portfolio overlap and common portfolio movement, but the design does not establish a causal effect.

4. Conclusion

This paper documents that diversification within institutional portfolios need not imply distinctiveness across institutional investors. Using SEC 13F holdings from 2018Q1 to 2025Q4, the evidence shows that managers with broader and less concentrated portfolios tend to hold portfolios that overlap more strongly with those of their peers. This relationship is consistent across alternative diversification measures and remains positive after excluding the largest institutional managers.

Portfolio crowding is also economically informative. Managers with more overlapping holdings exhibit stronger subsequent common portfolio movement with leave-one-out peer portfolios, including after accounting for broad market and factor exposures. These results suggest that conventional measures of portfolio diversification provide an incomplete view of institutional exposure because they capture concentration within a portfolio but not similarity across portfolios.

The analysis is descriptive and does not identify a causal effect of diversification on crowding or of crowding on returns. Nevertheless, the findings show that individually diversified portfolios can coexist with substantial collective overlap, making cross-manager similarity relevant for institutional portfolio monitoring and crowding-risk assessment.

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Appendix A. Additional descriptive and robustness evidence

Appendix Table A.1 reports manager-level summary statistics. Appendix Table A.2 reports price-universe sensitivity for the common portfolio movement evidence.

Table A.1: Summary statistics: 2018Q1–2025Q4

Variable	Mean	Std. Dev.	P25	Median	P75	N
Crowding mean	0.1652	0.1019	0.0796	0.1647	0.2457	16,000
Entropy	4.4990	1.4152	3.5975	4.7746	5.6089	16,000
HHI	0.0612	0.1457	0.0091	0.0173	0.0427	16,000
Effective N	77.85	74.10	23.39	57.86	110.41	16,000
Number of holdings	1071.7	1344.4	102.0	553.0	1519.0	16,000
Log portfolio value	27.3574	3.2287	24.3029	26.6477	30.1537	16,000
Max position	0.1151	0.1662	0.0395	0.0602	0.1039	16,000

Manager-level statistics for the top 500 13F managers by portfolio value each quarter. P25 and P75 denote the 25th and 75th percentiles.

Table A.2: Price-universe sensitivity for common portfolio movement

Priced-security universe	Focal variable	Coef.	SE	Observations
Top 100	Crowding mean	0.816***	(0.072)	6,282
Top 100	Abnormal crowding	0.800***	(0.074)	6,282
Top 300	Crowding mean	0.682***	(0.049)	6,395
Top 300	Abnormal crowding	0.672***	(0.052)	6,395
Top 500	Crowding mean	0.624***	(0.044)	6,427
Top 500	Abnormal crowding	0.614***	(0.046)	6,427

Raw common-correlation regressions using top- N priced securities each quarter. Peer returns are leave-one-out averages; residualized tests are diagnostic and more universe-sensitive.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.